Modeling a Planet Using Exoplex: Trappist-1e

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Motivation

1. The planet's interior structure and composition tells us about geological activity and mineral formation
2. Understanding a planet's habitability goes beyond its atmosphere.

Goal : Use stellar composition and planet mass/radius to model interior structure of Trappist-1e using ExoPlex.

Methods

Purpose

-Used Exoplex software to simulate planetary interiors

- Explored effects of varying Fe/Mg, Si/Mg, FeO in mantle, and light elements in the core

Example

- Modeled Trappist-1e using stellar data:
- Fe/Mg = 0.75
- Si/Mg = 1.02
- Mass = 0.772

Results

Modeled Radius

0.939 Earth radii

(observed: 0.92 ± 0.013)

Core mass Fraction

~27.7%

(of planet's total mass)

Central Pressure

117 GPa

Increased Si/Mg →

More garnet

5% FeO in mantle →

Higher Oxidation state

Likely Mineralogy:

More garnet, less olivine, possible perovskite/akimotoite

Conclusion

Phoenix-Avery Sarian - coding

Anthony Fauver- written report

Stella Nelson - presentation

- Trappist-1e is likely rocky and Earth-like in structure
- **Main differences:** lower core fraction, more oxidized mantle.
- Exoplex is an effective tool for approximating rocky planet interiors.
- **Future improvements:** refines stellar input and explore broader parameters.